



Artificial Intelligence

an introduction

Ahmad F. Al Musawi, PhD in AI

Department of Information Technology

College of Computer Science and Mathematics

University of Thi Qar, Iraq



AI Foundation

Lecture 1

Ahmad F. Al Musawi, PhD in AI

Department of Information Technology

College of Computer Science and Mathematics

University of Thi Qar, Iraq



AI foundation



- Content
 1. Learning Objectives
 2. Fundamental Definitions
 3. AI, ML, DL, and Generative AI
 4. Traditional AI
 5. How AI Learns
 6. Types of Machine Learning
 7. Major AI Capabilities
 8. AI Limitations
 9. Big Picture
 10. Bank of Questions for Lecture 1



AI foundation



Learning Objectives

Fundamental Definitions

AI, ML, DL, and Generative AI

Traditional AI

How AI Learns

Types of Machine Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions for Lecture 1

1. Learning Objectives

After this lecture, students should be able to:

- Define *Artificial Intelligence*.
- Explain the concept of *intelligence* in computational systems.
- Define an *intelligent agent*.
- Distinguish *AI*, *Machine Learning*, *Deep Learning*, and *Generative AI*.
- Distinguish *traditional programming* from *machine learning*.
- Describe *symbolic AI* and *machine-learning* approaches.
- Explain the concepts of *training*, *inference*, *prediction*, and *generation*.
- Identify major AI capabilities and application areas.
- Describe important milestones in AI history.
- Explain narrow AI and AGI.
- Identify major limitations of modern AI.

Learning Objectives

Fundamental Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

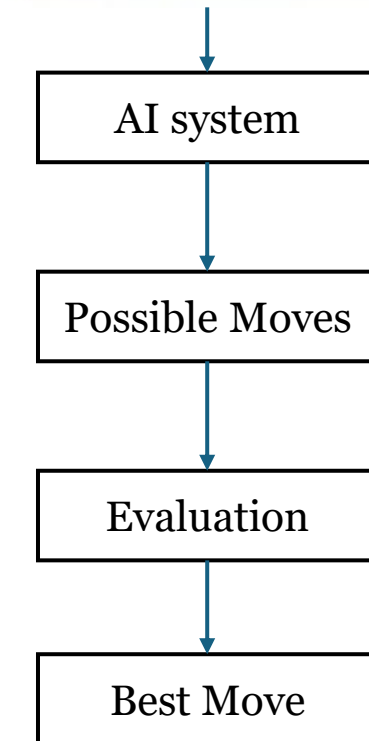
AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Fundamental Definitions

1. Artificial Intelligence
2. Intelligence
3. Intelligent agent
4. Rational agent



Learning Objectives

Fundamental Definitions

AI, ML, DL, and Generative AI

Traditional AI

How AI Learns

Types of Machine Learning

Major AI Capabilities

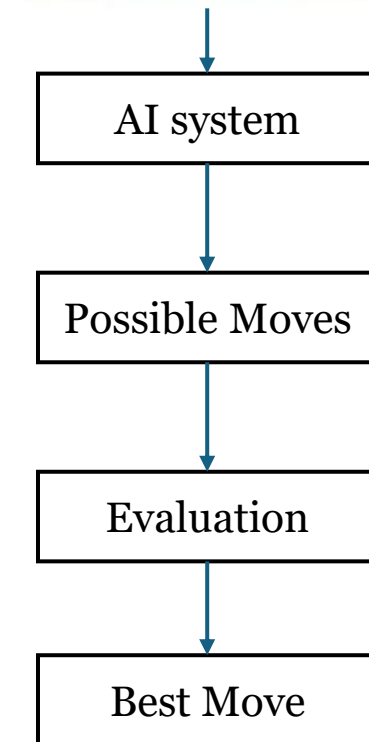
AI Limitations

Big Picture

Bank of Questions for Lecture 1

Artificial Intelligence

1. Definition **(AI)** is a field of computer science concerned with designing *computational systems* capable of *performing tasks* that normally require capabilities associated with human intelligence.
2. Capabilities
 1. Perception
 2. Learning
 3. Reasoning
 4. Problem solving
 5. Planning
 6. Decision making
 7. Language processing
 8. generation



Learning Objectives

Fundamental Definitions

AI, ML, DL, and Generative AI

Traditional AI

How AI Learns

Types of Machine Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions for Lecture 1

Intelligence

1. **Intelligence** is the ability to acquire and apply knowledge, learn from experience, reason about information, solve problems, adapt to changing situations, and make decisions toward a goal.
2. In AI, we are interested primarily in **computational manifestations of these capabilities**.
3. **Example:** The mechanisms are different, but both systems demonstrate goal-directed behavior.



See obstacle

Understand situation

Choose action

Avoid obstacle



sensors

perception

Decision model

Action

Learning Objectives

Fundamental Definitions

AI, ML, DL, and Generative AI

Traditional AI

How AI Learns

Types of Machine Learning

Major AI Capabilities

AI Limitations

Big Picture

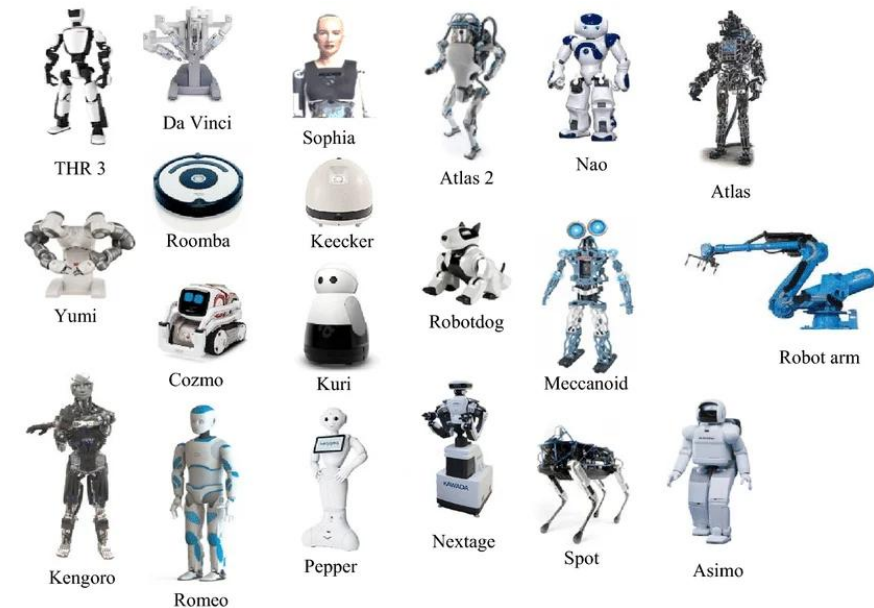
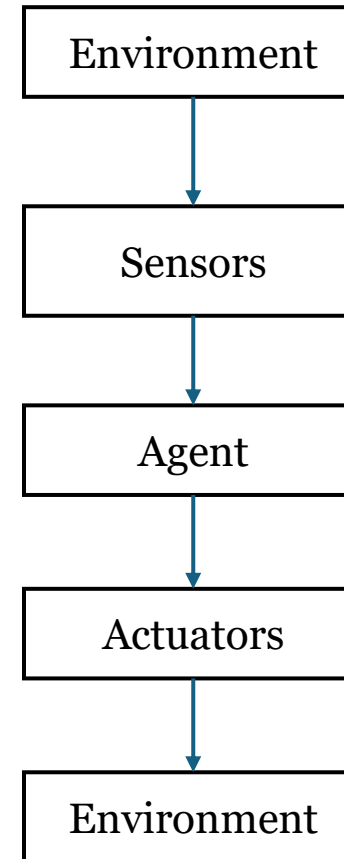
Bank of Questions for Lecture 1

Intelligent Agent

1. An **intelligent agent** is a computational system that perceives its environment and takes actions to achieve specified goals.

2. Example: Robot

1. Sensors: camera, microphone, distance sensor
2. Agent: decision making system
3. Actuator: wheels, robotic arm





Learning Objectives

Fundamental Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Rational Agent

- A **rational agent** selects actions expected to maximize its performance measure based on the information available to it.
- **Rational $\neq$ always correct.**
- An agent can make the rational decision given incomplete information and still obtain a bad outcome.

Learning Objectives

Fundamental
Definitions

**AI, ML, DL, and
Generative AI**

Traditional AI

How AI Learns

Types of Machine
Learning

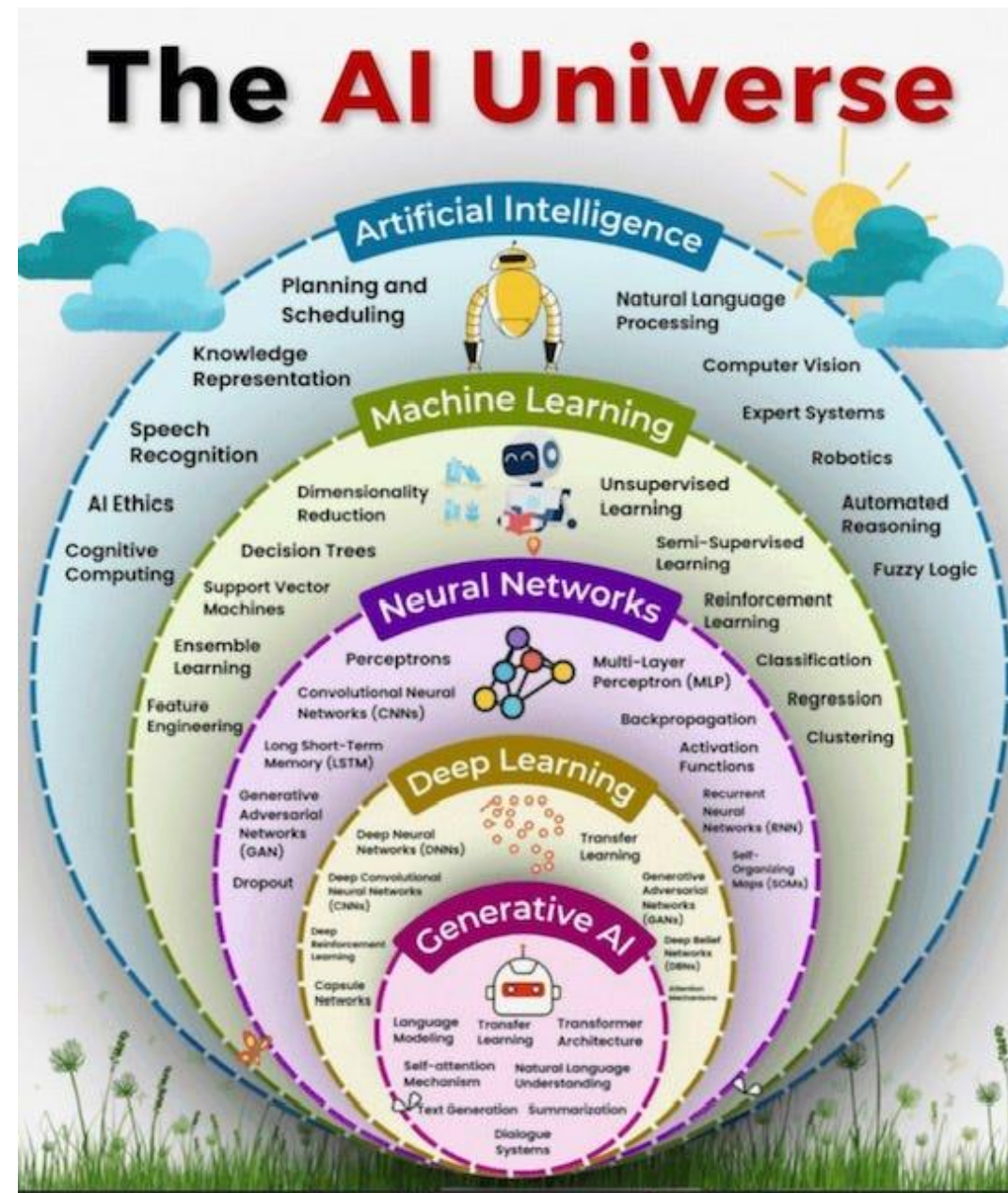
Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

- **AI,**
- **ML,**
- **DL and**
- **Generative AI**



Learning Objectives

Fundamental
Definitions

**AI, ML, DL, and
Generative AI**

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Machine Learning

- **Machine Learning (ML)** is a subfield of AI in which computational models learn patterns or relationships from data and use those learned patterns to make predictions, decisions, or other outputs.
- Figures 1, 2
- Example
 - Programming
 - If email contains “win money” : spam
 - ML
 - 10000 emails
 - Email 1 → spam
 - Email 2 → not spam
 - Email 3 → spam
 - ...

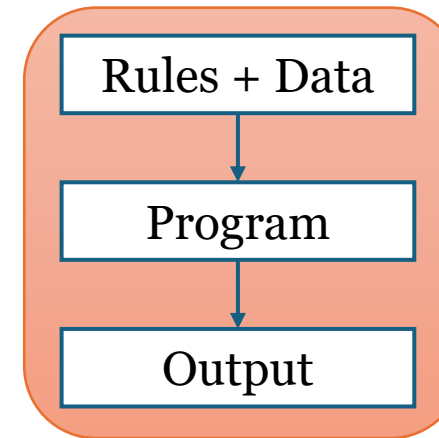


Fig 1. Traditional Programming

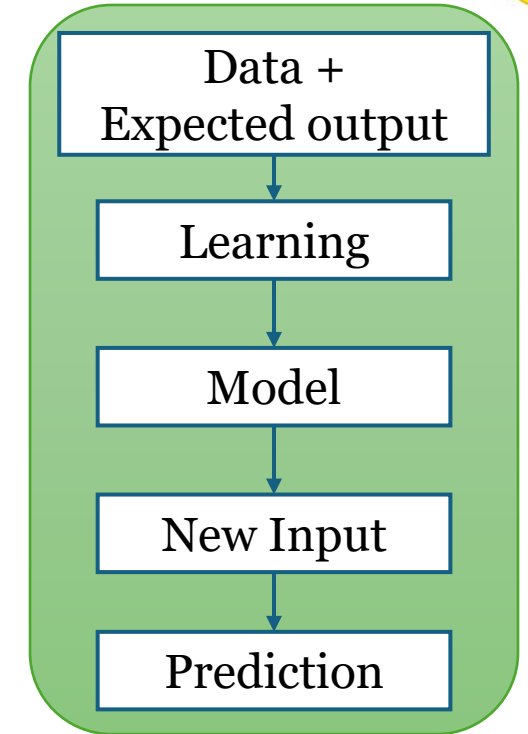


Fig 2. Machine Learning



Learning Objectives

Fundamental
Definitions

**AI, ML, DL, and
Generative AI**

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

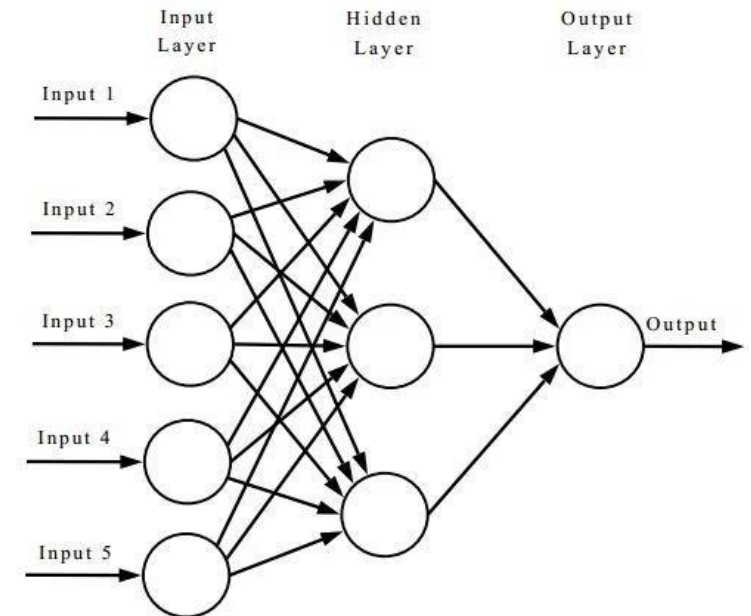
AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Deep Learning

- **Deep Learning (DL)** is a subfield of machine learning that uses neural networks with multiple computational layers to learn increasingly complex representations from data.
- Examples
 - CNN, Recurrent Neural Networks, Transform
- Deep?
 - Because the model contains **multiple layers** of learned transformations.



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

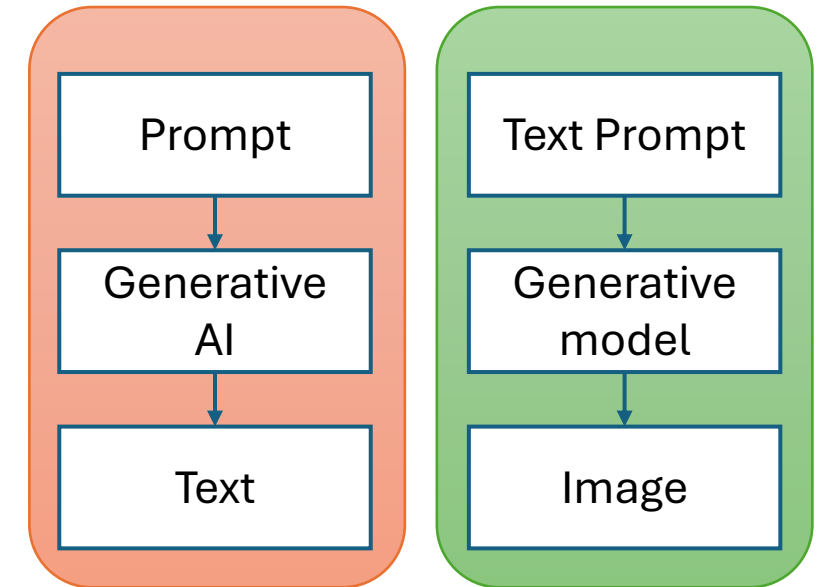
AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Generative AI

- **Generative Artificial Intelligence** refers to AI systems capable of generating new content such as text, images, audio, video, or code based on learned patterns.



- However, be aware of:
 - *Classification*:
 - "Is this image a cat or dog?" →
 - *Generation*:
 - "Generate an image of a cat playing violen."





Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Large Language Model (LLM)

- A **Large Language Model (LLM)** is a neural-network-based model trained on large amounts of text or other language data to model patterns in language and generate or process text.
- Examples of capabilities:
 - text generation
 - summarization
 - translation
 - question answering
 - code generation



Chatgpt



Claude



Gemini



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Traditional AI

- 1. Symbolic AI**
- 2. Expert systems**
- 3. Knowledge Representation**



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Symbolic AI

- **Symbolic AI** is an approach to artificial intelligence in which knowledge and reasoning are represented explicitly using symbols, rules, logic, and structured representations.
- **Example:**
IF fever AND cough
THEN possible respiratory infection
- **Characteristics**
 - explicit knowledge
 - rules
 - logic
 - reasoning
 - human-designed representations

Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

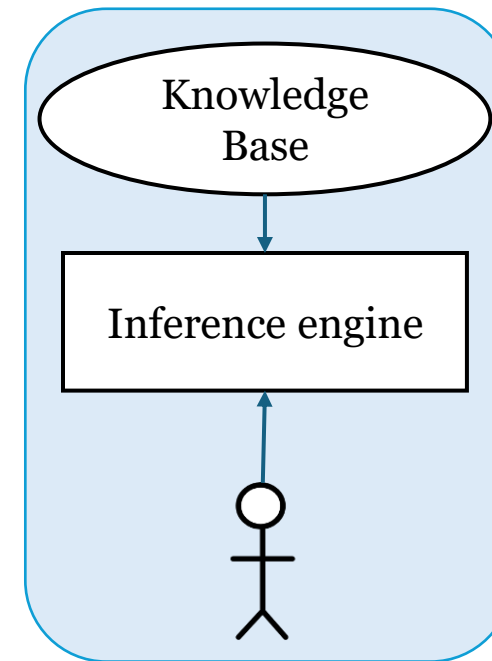
Big Picture

Bank of Questions
for Lecture 1

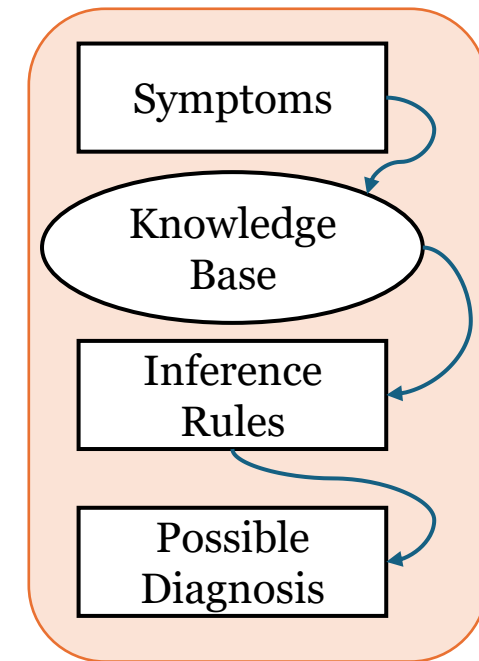
Expert System

- An **expert system** is an AI system that uses a knowledge base and an inference mechanism to solve problems in a particular domain.

Expert System



Example



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

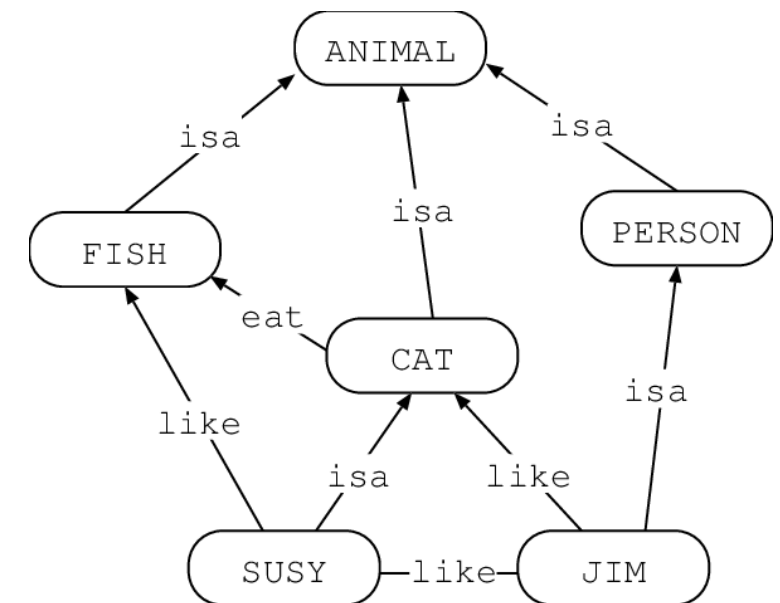
Big Picture

Bank of Questions
for Lecture 1

Knowledge Representation

- **Knowledge representation** is the process of encoding information about the world in a form that a computer can use for reasoning or decision making.

- **Examples:**
 - rules
 - logical statements
 - **semantic networks** →
 - ontologies
 - knowledge graphs





Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns

1. Data
2. Feature
3. Label
4. Model
5. Training
6. Inference
7. Prediction
8. Generalization



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Data

- **Data** are observations, measurements, records, or other information used by computational systems.
- Examples:
 - images
 - text
 - medical records
 - sensor measurements
 - audio
 - transactions



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Feature

- A **feature** is an individual measurable characteristic or input variable used by a machine-learning model.
- Example: For predicting *house* price:
 - area
 - bedrooms
 - bathrooms
 - age
 - location



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Label

- A **label** is the target value or category associated with an example in supervised learning.
- Example:
 - Patient data → Disease
- Here:
 - Age → feature
 - Blood pressure → feature
 - Glucose → feature
 - Disease → label



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

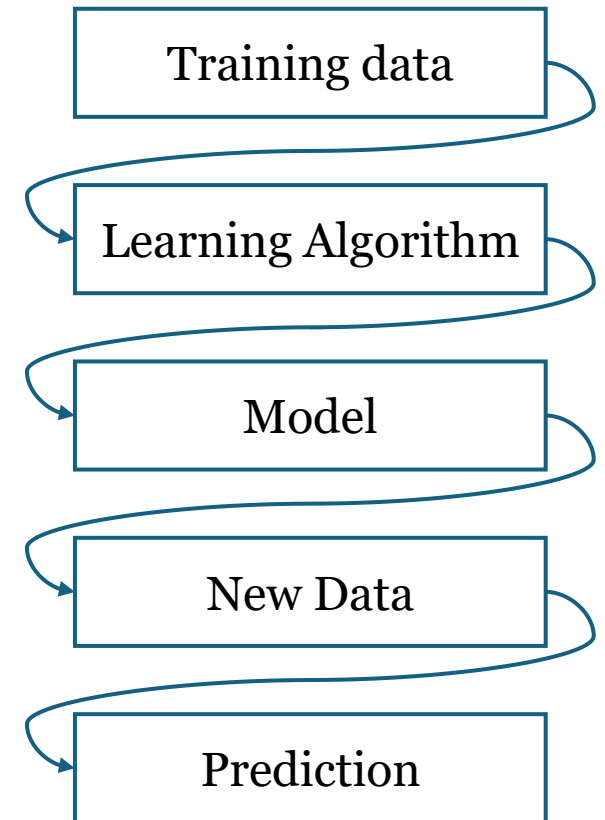
AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Model

- A **machine-learning model** is a learned mathematical representation of patterns or relationships in data that can be used to make predictions or decisions on new inputs.



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

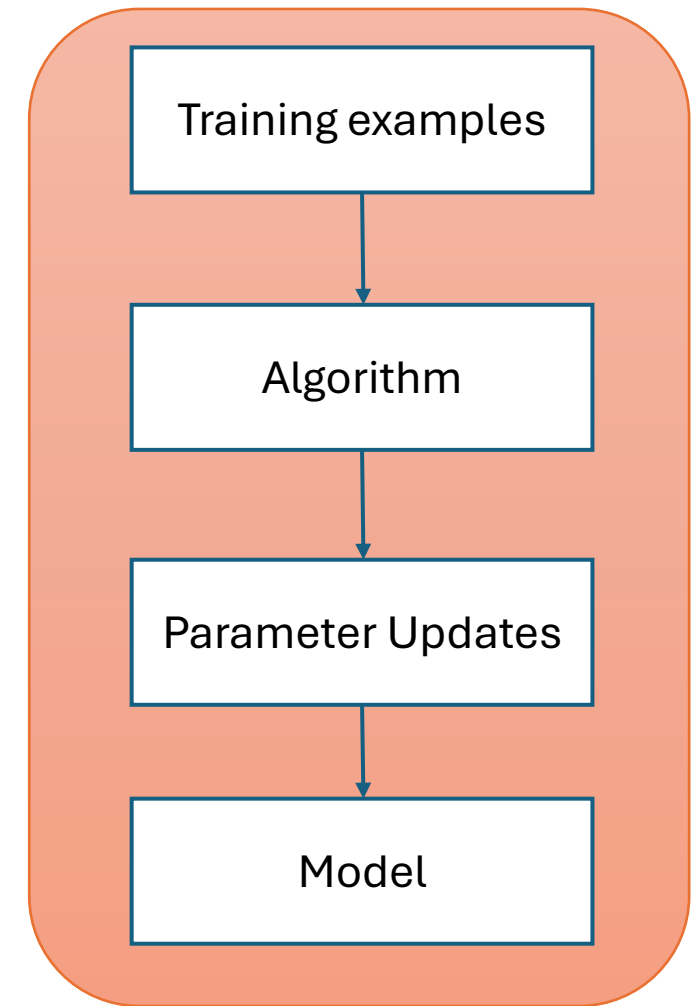
AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Training

- **Training** is the process of using data to adjust a model's parameters so that the model learns patterns relevant to a specified task.



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

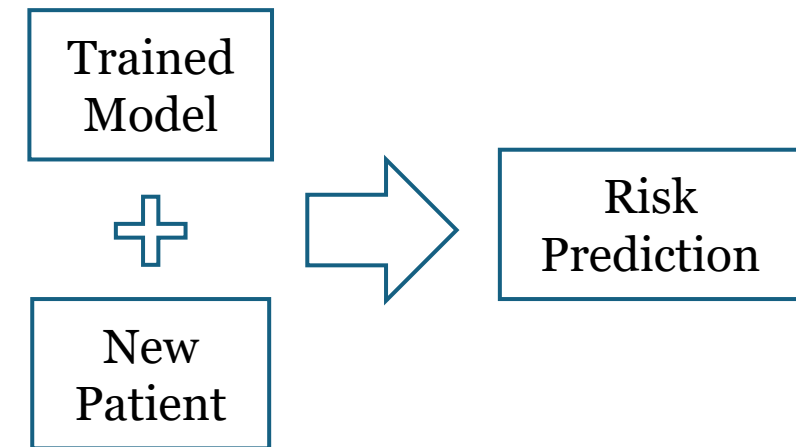
AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Inference

- **Inference** is the process of using a trained model to produce an output for new input data.
- Students should be able to distinguish:
 - **Training:** learning the model.
 - **Inference:** using the model.





Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Prediction

- A **prediction** is an output produced by an AI/ML model concerning an input or future/unknown value.
- Examples:
 - Patient $\rightarrow$ disease risk = 0.73
 - Image $\rightarrow$ cat
 - House $\rightarrow$ \$180,000



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

How AI Learns: Generalization

- **Generalization** is the ability of a trained machine-learning model to perform effectively on previously unseen data.
- Example:
 - A model memorizes training examples:
 - Training accuracy = 99%
 - Test accuracy = 55%
 - Poor generalization.
- A useful model should learn patterns rather than simply memorize examples.



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

**Types of Machine
Learning**

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Types of Machine Learning

- 1. Supervised Learning**
- 2. Unsupervised Learning**
- 3. Reinforcement learning**



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

**Types of Machine
Learning**

Major AI Capabilities

AI Limitations

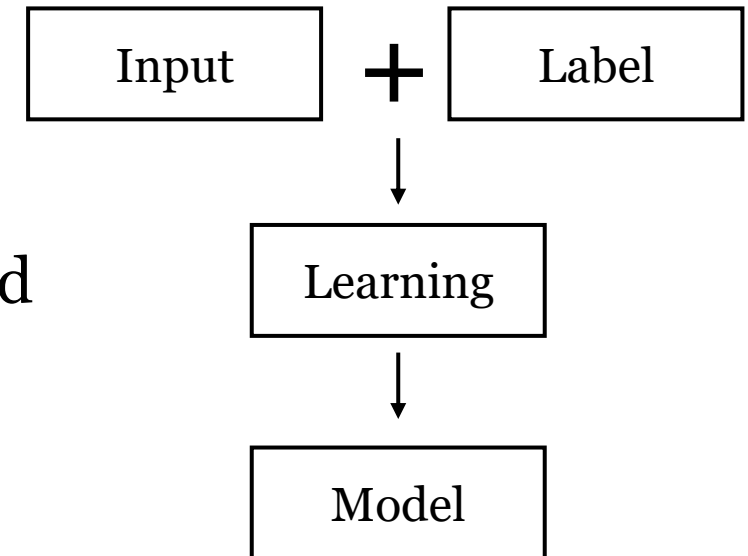
Big Picture

Bank of Questions
for Lecture 1

Types of Machine Learning

1. Supervised Learning

- **Supervised learning** is machine learning in which a model learns from labeled examples consisting of inputs and corresponding target outputs.
- Examples:
 - disease classification
 - spam detection
 - house-price prediction





Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

**Types of Machine
Learning**

Major AI Capabilities

AI Limitations

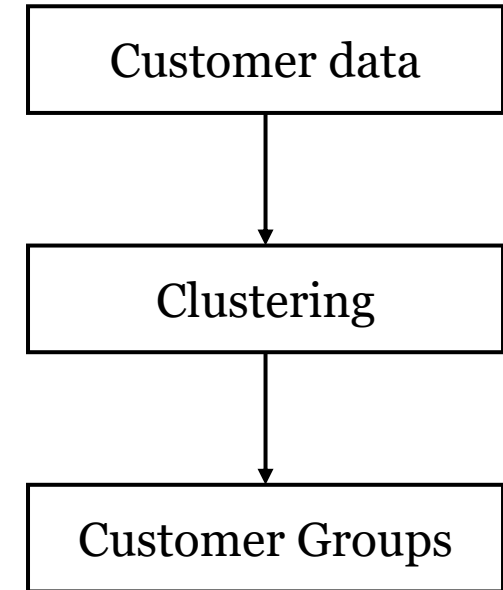
Big Picture

Bank of Questions
for Lecture 1

Types of Machine Learning

2. Unsupervised Learning

Unsupervised learning is machine learning in which a model learns *patterns* or *structures* from data *without* predefined target *labels*.





Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

**Types of Machine
Learning**

Major AI Capabilities

AI Limitations

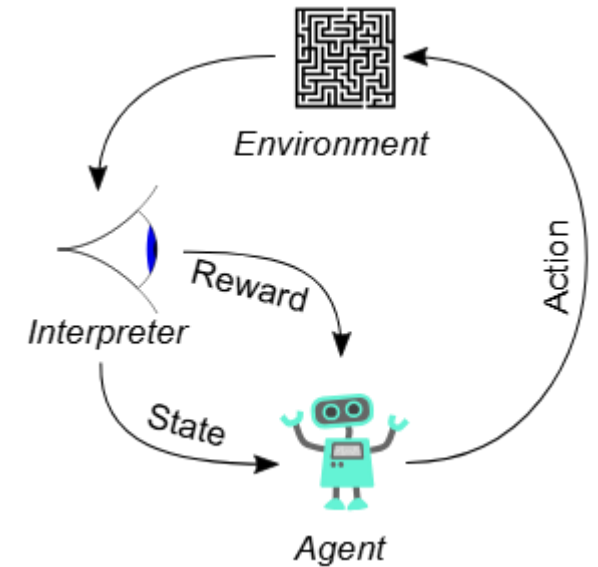
Big Picture

Bank of Questions
for Lecture 1

Types of Machine Learning

3. Reinforcement learning

Reinforcement learning (RL) is a machine-learning paradigm in which an agent learns by interacting with an environment and receiving rewards or penalties associated with its actions.





AI foundation



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Major AI capabilities

1. Perception
2. Natural Language Processing
3. Computer Vision
4. Reasoning
5. Planning

Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

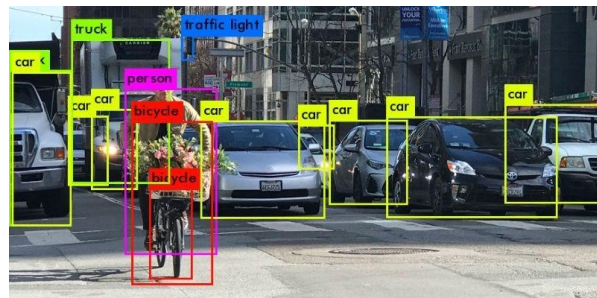
Big Picture

Bank of Questions
for Lecture 1

Major AI capabilities

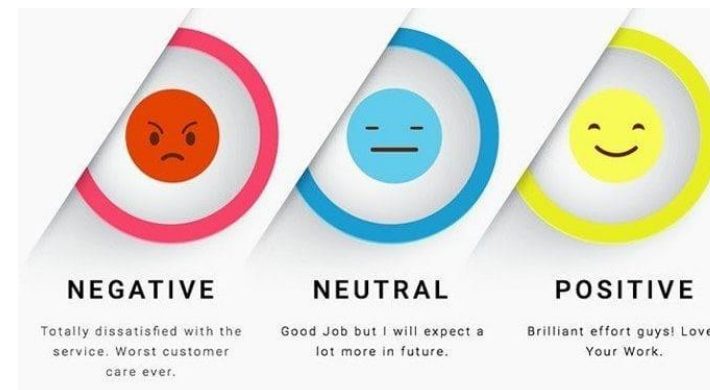
1. Perception is the ability of an AI system to obtain and interpret information from sensory or observational data.

- Examples:
 - **image recognition**
 - speech recognition
 - object detection



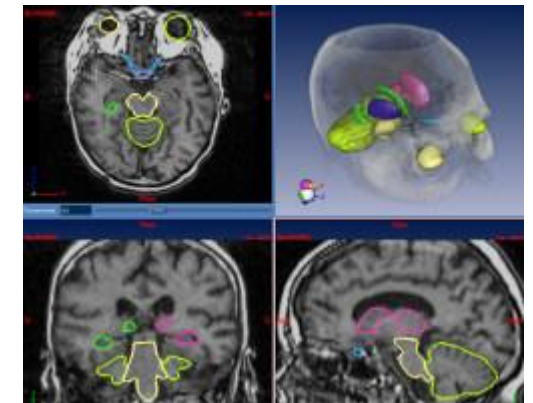
2. Natural Language Processing (NLP) is the area of AI concerned with computational processing, understanding, and generation of human language.

- Examples:
 - Translation
 - **sentiment analysis**
 - summarization
 - question answering



3. Computer Vision is the area of AI concerned with enabling computers to analyze and interpret visual information.

- Examples:
 - image classification
 - object detection
 - face recognition
 - **medical image analysis**





Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

Major AI capabilities

4. AI reasoning refers to the process of deriving conclusions or selecting actions based on available information, knowledge, or learned representations.

- Examples:
 - logical inference
 - diagnosis
 - planning

5. Planning is the process of determining a sequence of actions intended to achieve a specified goal.





AI foundation



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions
for Lecture 1

AI Limitations

1. Bias
2. Hallucination
3. Overfitting
4. Explainability



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions

AI Limitations

- 1. AI bias** refers to systematic patterns in an AI system's outputs that can produce unfair or undesirable results, often associated with characteristics of the data, problem formulation, or system design.
- 2. AI hallucination** refers to an AI system generating information that appears plausible but is unsupported, incorrect, or fabricated.



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions

AI Limitations

3. Overfitting occurs when a model learns the training data too closely, including noise or incidental patterns, resulting in poor performance on unseen data. For example :

- Training performance → Excellent
- Unseen-data performance → Poor

4. Explainability concerns the extent to which humans can understand how or why an AI system produced a particular output. Important in:

- healthcare
- finance
- law
- safety-critical systems

Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions

Big Picture

Artificial Intelligence

Symbolic AI

Logic

Search

Rules

Machine Learning

Supervised
Learning

Unsupervised
Learning

Reinforcement
Learning

Deep Learning

CNN

Transformers

GNN

LLM

Generative
AI



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

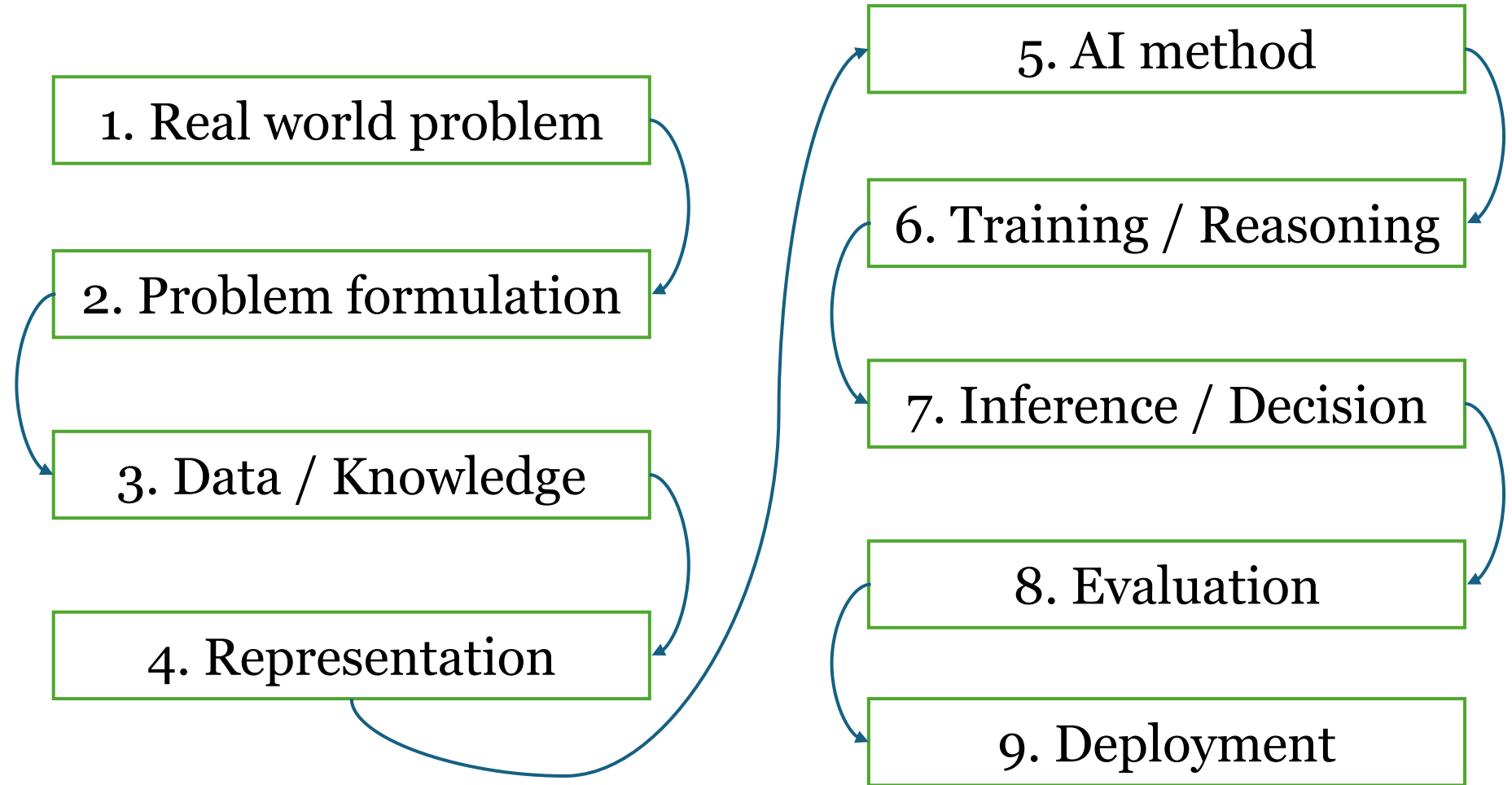
Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions

AI Problem solving pipeline





Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions

Bank of Questions

Definition questions

1. Define Artificial Intelligence.
2. Define Machine Learning.
3. Define Deep Learning.
4. Define Generative AI.
5. Define an intelligent agent.
6. Define a rational agent.
7. Define supervised learning.
8. Define unsupervised learning.
9. Define reinforcement learning.
10. Define an AI model.
11. Define training.
12. Define inference.
13. Define generalization.
14. Define overfitting.
15. Define NLP.
16. Define computer vision.
17. Define symbolic AI.
18. Define an expert system.

Explain questions

19. Explain how an intelligent agent interacts with its environment.
20. Explain the difference between training and inference.
21. Explain how machine learning differs from traditional programming.
22. Explain the concept of generalization.
23. Explain how an expert system works.
24. Explain why AI is broader than machine learning.
25. Explain why deep learning is considered a subset of machine learning.

Application questions

26. Given a medical prediction problem, identify: features, labels, model, training, inference.



Learning Objectives

Fundamental
Definitions

AI, ML, DL, and
Generative AI

Traditional AI

How AI Learns

Types of Machine
Learning

Major AI Capabilities

AI Limitations

Big Picture

Bank of Questions

Bank of Questions

1. Why does high training accuracy not necessarily indicate a good AI model?
2. Can a system be considered AI if it does not use machine learning? Explain.
3. Can a machine-learning system make incorrect predictions even when it was correctly trained? Explain.
4. Why is data quality important in AI?
5. Why should AI outputs not automatically be considered correct?

Questions?

